

AUTOMATIC SOIL MOISTURE DETECTION AND WATER PUMPING SYSTEM

Arduino Uno Based Project Report

Project Description

This project is an Arduino-based Automatic Soil Moisture Detection and Water Pumping System. An Arduino Uno is used as the main controller, while a potentiometer is used in the simulation to represent the output of a soil moisture sensor. Arduino reads the analog value, compares it with a predefined threshold, and automatically controls a DC motor/pump. When the soil is detected as dry, the pump turns ON; when sufficient moisture is detected, the pump turns OFF.

1. Introduction

The Automatic Soil Moisture Detection and Water Pumping System is designed to automatically control a water pump according to the moisture condition of soil. It reduces unnecessary watering and human effort. In the simulation, the potentiometer represents a soil moisture sensor and allows the sensor value to be varied manually.

2. Objectives of the Project

1. Detect the moisture condition of soil.
2. Automatically control a water pump.
3. Reduce unnecessary water consumption.
4. Reduce human effort in watering.
5. Demonstrate Arduino analog input and digital motor control.
6. Develop a simple and low-cost automatic irrigation system.

3. Components Required

Arduino Uno, potentiometer, DC motor/pump, NPN transistor, diode, resistor, jumper wires, suitable power supply, and USB cable. In a practical system, the potentiometer should be replaced by an actual soil moisture sensor.

4. Block Diagram

Soil Moisture Sensor / Potentiometer → Arduino Uno → Decision Based on Threshold → Motor Control Circuit → DC Water Pump → Water Supply to Soil

5. Working Principle

The potentiometer is connected to an analog input of Arduino. Its changing voltage represents different soil moisture conditions. Arduino's ADC converts the analog voltage into a value from 0 to 1023. The program uses a threshold of 500. If the sensor value is below 500, Arduino considers the soil dry and switches the pump ON. If the value is 500 or more, the pump is switched OFF.

6. Circuit Description

Arduino Uno: Main controller that reads the sensor and controls the motor.

Potentiometer: Used in the simulation to imitate a soil moisture sensor. One outer terminal is connected to 5V, the other to GND, and the middle terminal to an analog input such as A0.

DC Motor/Pump: Represents the water pump and is controlled through a transistor switching circuit.

Transistor: Works as an electronic switch because an Arduino GPIO pin should not directly drive a motor.

Diode: Protects the motor-driving circuit from reverse voltage/back EMF produced when the motor is switched OFF.

Resistor: Limits the transistor base current.

7. Pin Configuration

Arduino Pin	Connected Component	Function
A0	Potentiometer / Sensor	Analog sensor input
D8	Motor control circuit	Turns pump ON/OFF
5V	Potentiometer	Sensor supply
GND	Potentiometer & circuit	Common ground

8. Software Used

The project can be developed and tested using the Arduino IDE or a simulator such as Tinkercad. The program is written in Arduino C/C++. The Serial Monitor is used to display sensor readings and pump status.

9. Program Logic / Algorithm

START → Initialize pins → Read sensor value → Compare with threshold → If value < 500, Motor ON → Otherwise Motor OFF → Display result on Serial Monitor → Wait 500 ms → Repeat.

10. Arduino Code

```
const int sensorPin = A0;
const int motorPin = 8;
const int threshold = 500;

void setup() {
  pinMode(motorPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(sensorPin);

  Serial.print("Sensor Value: ");
  Serial.println(sensorValue);

  if (sensorValue < threshold) {
    digitalWrite(motorPin, HIGH);
    Serial.println("Soil Dry - Pump ON");
  }
  else {
    digitalWrite(motorPin, LOW);
    Serial.println("Soil Wet - Pump OFF");
  }

  delay(500);
}
```

11. Explanation of Important Code

sensorPin = A0: Defines the analog sensor input.

motorPin = 8: Defines Arduino digital pin 8 for motor control.

threshold = 500: Sets the reference value for dry/wet decision.

analogRead(): Reads the analog sensor value from 0 to 1023.

digitalWrite(HIGH): Turns the motor control signal ON.
digitalWrite(LOW): Turns the motor control signal OFF.
Serial.begin(9600): Starts serial communication at 9600 baud.
delay(500): Waits 500 milliseconds before the next reading.

12. Threshold Concept

Sensor Value	Condition	Motor
Less than 500	Soil Dry	ON
500 or more	Soil Wet	OFF

The threshold can be calibrated according to the actual sensor and soil type. For example, the value 500 can be changed in the program if testing shows that another reference value provides better results.

13. Role of Potentiometer in the Simulation

The potentiometer does not actually measure soil moisture. It is used as a convenient simulation substitute for a soil moisture sensor. Turning the potentiometer changes the analog voltage and therefore changes the value read by Arduino. A value below 500 represents dry soil in this project, while 500 or above represents sufficiently wet soil.

14. Role of Transistor

A DC motor normally requires more current than an Arduino GPIO pin should supply directly. The transistor therefore acts as an electronic switch. When Arduino pin 8 outputs HIGH, the transistor switches ON and the motor runs. When the pin outputs LOW, the transistor switches OFF and the motor stops. A suitable external motor supply should be used in a practical circuit.

15. Role of Diode

A DC motor is an inductive load. When it is switched OFF, it can produce a reverse voltage called back EMF. The diode provides a safe path for this voltage and helps protect the transistor and Arduino-side control circuit.

16. Advantages

1. Automatic operation.
2. Saves water by supplying it only when required.
3. Low-cost components.
4. Easy to implement and understand.
5. Reduces manual watering effort.
6. Real-time sensor monitoring through Serial Monitor.
7. Can be expanded with additional sensors and IoT features.

17. Limitations

1. The potentiometer is only a simulation of a moisture sensor.
2. A real moisture sensor is required for practical implementation.
3. Threshold calibration may be needed for different soil types.
4. The motor needs an appropriate external power supply.
5. The present system does not measure water quantity.
6. It does not detect whether the water tank is empty.

18. Applications

Home gardens, agricultural fields, greenhouses, plant nurseries, indoor plants, automatic irrigation systems, smart farming, and small-scale farming.

19. Future Scope

The system can be improved by replacing the potentiometer with a real capacitive or resistive soil moisture sensor, adding a water-level sensor, LCD display, Wi-Fi/IoT connectivity, multiple moisture sensors, and time-based irrigation control.

20. Expected Output

When the sensor value is below the threshold:

Sensor Value: 400

Soil Dry - Pump ON

When the sensor value is 500 or above:

Sensor Value: 700

Soil Wet - Pump OFF

21. Conclusion

The Automatic Soil Moisture Detection and Water Pumping System successfully demonstrates sensor-based automatic irrigation using Arduino Uno. In the simulation, a potentiometer represents the soil moisture sensor. Arduino reads its analog value and compares it with a threshold of 500. If the soil is considered dry, Arduino activates the motor through a transistor switching circuit; when sufficient moisture is detected, the motor is switched OFF. The project demonstrates Arduino programming, analog signal processing, automatic decision-making, and motor control, and it can be expanded into a practical smart irrigation system.

Project Type: Arduino / Embedded Systems / Automatic Irrigation